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Security and Privacy Requirements for RFID Schemes in Healthcare: Case Studies, Solutions, and Challenges

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Abstract

RFID technology has proven very useful in many healthcare applications, starting with medical asset tracking, continuing with staff tracking, and reaching medical event and procedure tracking. Security and privacy properties play a fundamental role in this range of applications. The protocols proposed so far highlights many techniques for analyzing these properties, making it difficult to compare the protocols with each other most of the time. In the case of lightweight protocols, the comparisons are even more challenging to achieve because these resort to idealizations that usually have no practical support, such as protocols based on physically unclonable functions (PUFs), or because no security and privacy model is known so far. This paper draws attention to the HPVP model as a possibility for uniformly studying healthcare RFID protocols' security and privacy properties. We highlight two general patterns that, once encountered in RFID protocols, show that they do not satisfy privacy or mutual authentication requirements. We then illustrate these templates through several proposed RFID protocols for use in healthcare. Then, we propose a classification of healthcare services based on an extended hierarchy of HPVP privacy classes, and we highlight RFID protocols that reach the respective levels of privacy.

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1. Introduction

Radio frequency identification (RFID) has become extremely useful in identification and authentication in the last 20 years. The areas covered are increasingly vast, and the usefulness of RFID technology in healthcare is truly spectacular. Starting from medical asset tracking and continuing with medical staff, event, and process tracking, RFID-based identification irrevocably proves its usefulness in healthcare. InsightAce Analytics recently showed in their Report ID 1419 that RFID in the healthcare market size, valued at USD 6.29 billion in 2023, is predicted to reach USD 20.28 billion by 2031.

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RFID authentication protocols are fundamental components of the RFID technology. As a result, it is unsurprising that in the last 10-15 years, many such protocols have been proposed, as diverse as possible, due to the nature of the scientific support used. We thus find protocols based on lightweight cryptography, symmetric cryptography, and various flavors of public key cryptography (based on number theory, elliptic curves, codes, lattices, etc.). Physically unclonable functions (PUF) technology was also incorporated into this framework.

The security and privacy properties an RFID protocol should offer play a crucial role. These properties are studied mainly through ad hoc, informal techniques or techniques not explicitly developed for RFID. This makes it challenging to compare RFID protocols because a specific property offered by a protocol and analyzed according to a particular method may not be valid in the case of another protocol that provides a similar property but is studied according to another technique. But let us look into the literature dedicated to RFID protocols. In the last 17 years, several general models for analyzing security and privacy properties have been consolidated and matured. We mention Vaudenay's [15] and the HPVP [6] models, which offer a suitable hierarchy of privacy properties. Unfortunately, not many RFID protocol proposals study the security and privacy properties through these models. In addition, the analysis of many proposed protocols shows us that they do not even reach the lowest level of privacy in these models (our Section 4 illustrates this).

Contributions and Paper Structure. Our paper starts from the premise that using a reputable and mature model to study security and privacy properties for RFID protocols can significantly consolidate the RFID technology in healthcare. Healthcare services have different degrees of privacy, and as a result, the protocols that serve them can be lightweight when the privacy requirement is not very high or incorporate more consistent cryptographic primitives when the privacy requirement is high. In light of these, we adopt the HPVP security and privacy model in RFID (Section 2) and do the following:

1. We argue that the HPVP model for RFID security and privacy is suitable for the healthcare field (Section 3.1);
2. We add one more privacy level, namely *r-weak*, to the four privacy levels *weak*, *forward*, *destructive*, and *strong* offered by the HPVP model (Section 3.2);
3. We identify healthcare services whose privacy falls on the mentioned levels (Section 3.2);
4. We highlight RFID protocols that provide privacy for each of the privacy levels mentioned above (Section 5);
5. We highlight RFID protocols proposed in specialized literature, which either do not reach any level of privacy in the HPVP model or do not ensure mutual authentication (Section 4).

The last section concludes the paper.

2. Basics of RFID Technology

In this section, we will present some basic elements of RFID technology, which we will use later. Before, however, we fix some standard cryptography notations (for details, the reader is directed to [9]).

We use in our exposition *probabilistic polynomial time* (PPT) algorithms $\mathcal{A}$ as defined in [14] that can consult *oracles*. For a set A , $a \leftarrow A$ means that a is uniformly at random chosen from A . If $\mathcal{A}$ is a probabilistic algorithm, then $a \leftarrow \mathcal{A}$ means that a is an output of $\mathcal{A}$ for some given input.

An *RFID system* is typically composed of three main entities [15]: a *reader*, a set of *tags*, and a radio frequency *communication protocol* between reader and tags. The reader is a powerful device not computationally restricted so it can perform any cryptographic operation. It stores tag related information in a database to which it has secure access. On the other side, tags are small devices that are considered to be resource constrained. The memory of a tag is typically split into *permanent* (or *internal*), used to store the state values of the tag, and *temporary* (or *volatile*), used to carry out the calculations required by the communication protocol. There are two types of temporary variables, *local*, used by tags only to do computations in a given protocol step, and *global*, that get values in a given protocol step and are used in another protocol step.

The communication protocol is an alternating sequence of reader-to-tag and tag-to-reader communication steps in which the first step can be taken by either of them. When the reader sends a message m to the tag, we will often say that the reader queries the tag on m . When the first protocol step is taken by the tag, we will say that the tag answers to the empty query (this corresponds to the tag being powered by the reader).

The *correctness* of an RFID scheme refers to the honest behavior of the reader and tag in a complete protocol session. That is, regardless of how the system is set up, after each complete execution of the interactive protocol between the reader and a legitimate tag, the reader outputs tag's identity with overwhelming probability. For mutual authentication RFID schemes, *correctness* means that the reader outputs tag's identity and the tag outputs *OK* with overwhelming probability.

The newest technologies allow RFID systems with tags equipped with *physically unclonable functions* (PUFs) [11]. A PUF can be seen as a physical object that, when queried with a challenge x generates a response y that depends on both x and the specific physical properties of the object. PUFs are typically assumed to be *physically unclonable* (it is infeasible to produce two PUFs that cannot be distinguished based on their challenge/response behavior), *unpredictable* (it is infeasible to predict the response to an unknown challenge), and *tamper-evident* (any attempt to physically access the PUF irreversibly changes the challenge/response behavior).

Tags equipped with PUFs will be called *PUF tags*. An RFID scheme with PUF tags will sometimes be called *PUF-based RFID scheme*. The main advantage in using PUF tags is that the corruption on a PUF tag reveals the permanent (and temporary) memory of the tag, but the tag is considered destroyed. By corruption, the values computed by PUFs cannot be obtained (except when they were saved in the permanent memory or in non-local temporary variables).

The HPVP security and privacy model [6] allows an adversary (a PPT algorithm) to query the following oracles:

- *CreateReader()*: Creates a new reader, and a unique reference R to it is returned;
- *CreateTag(ID)*: Creates a tag with the identifier ID by calling $SetupT(pk, ID)$. The tag is registered in the server's database (that is, the oracle creates only legitimate tags). Moreover, duplicate tags with the same ID are accepted. A unique reference T to the tag is returned;
- *RegisterTag(T, R)*: Registers the tag T with the reader R ;
- *Launch(R)*: Generates and outputs a new protocol session identifier π with the reader R ;
- *DrawTag(T_0, T_1)*: Generates a fresh virtual tag reference $vtag$ that refers to either the left tag T_0 or to the right tag T_1 , depending on the privacy game where the oracle is queried. The triple $(vtag, T_0, T_1)$ is included in a list Γ of drawn tags, and $vtag$ is returned by the oracle. The oracle returns $\perp$ if one of the two tags is in the insider list, or one of the two tags is registered with a different set of readers than the other tag, or T_0 (T_1) is already referenced as the left-hand (right-hand) side tag in Γ ;
- *Free(vtag)*: Resets (erases) the temporary state of the tag referenced by $vtag$ and removes the corresponding triple from Γ ;
- *SendTag($m, vtag$)*: Outputs the tag's answer when the message m is sent to the tag referred to by $vtag$. When m is the empty message, this oracle outputs the first message of the protocol instance π , assuming that the tag does the first step in the protocol;
- *SendReader(R, m, π)*: Outputs the R 's reader answer when the message m is sent to it as part of the protocol instance π . When m is the empty message, abusively but suggestively denoted by $\emptyset$, this oracle outputs the first message of the protocol instance π , assuming that the reader does the first step in the protocol;
- *Result(π)*: Outputs $\perp$ if in session π the reader has not yet made a decision on tag authentication (this also includes the case when the session π does not exist), 1 if in session π the reader authenticated the tag, and 0 otherwise (this oracle is both for unilateral and mutual authentication);
- *Corrupt(T)*: Outputs the full (permanent and temporary) state of the tag T . Remark that the corruption is with respect to a tag, not a virtual tag. Otherwise, it would be easy for an adversary with corruption capabilities to distinguish between the left and the right privacy games [5, 6];
- *CreateInsider(ID)*: Creates a tag and returns a unique reference T to it and its full state. The tag is included in a list of insider tags. This oracle is a method of giving the adversary the internal state of a tag but without considering that the adversary has used the corruption oracle.

In the HPVP model, the adversary can be classified as follows:

1. Adversaries with no access to *CreateInsider*. These are further classified according to the way the *Corrupt* oracle is used: *weak adversaries* (no access to *Corrupt*), *forward adversaries* (once they access the *Corrupt* oracle, the only oracle they can access is *Corrupt*), *destructive adversaries* (the tag is destroyed after corruption), *strong adversaries* (no restrictions), *narrow adversaries* (no access to *Result*);

2. Adversaries with access to *CreateInsider*. The power of a destructive or strong adversary does not increase if he is given access to the *CreateInsider* oracle.

Security in the HPVP model means that no strong adversary has more than a negligible probability to make the reader authenticate an uncorrupted legitimate tag without having any tag authentication matching conversation. When the RFID scheme is with mutual authentication, besides the above requirement, it is asked that no strong adversary has more than a negligible probability to make an uncorrupted legitimate tag to authenticate the reader without having any reader authentication matching conversation. Privacy in the HPVP model generalizes well-known properties such as anonymity, unlinkability, etc. An RFID scheme is considered *C-private*, where *C* is a class of adversaries, if no adversary in *C*, after consulting the oracles, can distinguish with more than negligible probability with which tag he communicated (the left or the right in the pairs drawn by him). The previously defined adversary classes lead to a ranking of the privacy properties of RFID schemes as shown in the diagram in Figure 1.

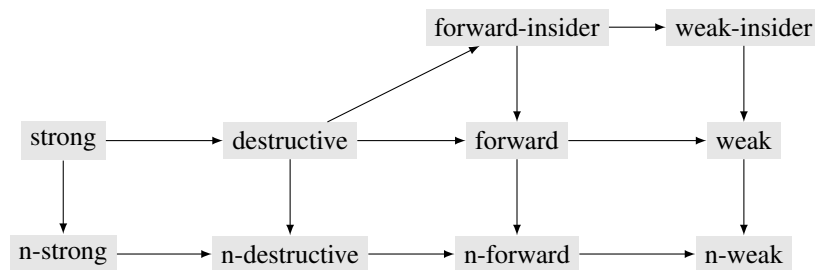


Fig. 1. Privacy levels in the HPVP model: “n-p” means “narrow p” and an arrow means “implication”.

3. The RFID Technology in Healthcare

The primary purpose of RFID technology is to identify and authenticate certain entities. Together with these, data can be collected about the entities in question, which can then be processed to extract useful information. Healthcare par excellence is a field in which identification and authentication play an important role. Not only that, in this field, we deal with great mobility of the entities that need to be identified (patients, analyses, medical devices, medicines, etc.), but we also have to deal with major security and privacy requirements (of medical data, procedures applied in various situations, devices that must keep patients alive, etc.). Thus, it is not surprising that healthcare has been embracing RFID technology for some time. Together with cloud and IoT technology, it brings significant benefits to securing streamlined identification, care history, traceability, equipment management (including inventory and maintenance), and traceability of blood bags, organs, samples, and many more.

3.1. Healthcare Solutions Using RFID

The multitude of healthcare applications that RFID technology can contribute can be classified as follows.

Medical asset tracking. Many technologies, such as Bluetooth Low Energy (BLE) trackers, barcodes, and QR codes, can be used for asset tracking. However, when it comes to medical needs, RFID technology has proven to be one of the most efficient options. This technology can provide a number of essential features, such as:

- Asset requests and real-time location tracking;
- Medical asset data storage;
- Analytics and reporting on asset utilization;
- Inventory quality control and maintenance;
- Asset security monitoring.

The medical assets that can be tracked can be classified into:

- *Medical tools*, such as disposable supplies (e.g., boxes of gloves, packs of syringes) or reusable medical tools (e.g., scalpel, scissors, retractors). The RFID technology can be used to locate them and track vital information on their use (e.g., expiration dates, data on sterilization);
- *Medications*. The RFID technology can help get information on the location, quantity, and expiration dates of medication;
- *Medical devices and equipment*, such as diagnostic medical devices, life-support equipment (e.g., artificial lung ventilation machines, kidney dialysis machines), or personal protection equipment (PPE). RFID technology can greatly help in this case by providing location information, length of use, maintenance schedule, etc.

People tracking. In a healthcare system, we usually meet three categories of people: patients, staff, and visitors. Each category has separate specifications that can vary even within the category. For example, neonatal, pediatric, mental health, or geriatric unit patients have different specifications. In general, the specifications refer to *identification*, *localization*, and *safety*. They can also refer to the *rights* and *permissions* that people from certain categories have. For example, a doctor can access certain areas of a hospital, certain medical equipment, or drugs. In such situations, the RFID system must provide certain authentication guarantees to prevent patients from accessing areas not allowed to them, access to medical equipment with severe restrictions on use, or drug theft.

The patient safety requirements are very varied. For example, RFID technology can be used to monitor implants. To ensure patient safety and enable fast and efficient product recalls, medical regulations require that each surgical implant be identifiable by the batch number and even uniquely by its serial number.

Event and procedure tracking. RFID systems can also monitor healthcare events, activities, and procedures. This can be very helpful in the medication process for patients or their behavior to ensure their safety. The continuous monitoring of various parameters (temperature, fluid flow, etc.) helps to streamline the workflow. Also, signaling the need to perform specific procedures or steps in a procedure can prevent malfunctions in the medical act.

Obviously, in such situations, the RFID system must be subjected to consistent verification of its proper functioning, and its security must be guaranteed (no entity with bad intentions can modify its proper functioning). Also, privacy needs can be at a high level.

3.2. Security and Privacy Requirements

As we have already mentioned, the paper's motivation starts from the fact that many RFID protocols have been proposed for healthcare over time. In many cases, their security and privacy analysis was carried out ad hoc or informally, which often was wrong. In some cases, the protocols were fixed through protocols that later proved vulnerable. On the other hand, we already have pretty mature RFID security and privacy models that can be used to analyze the proposed protocols. In addition, using these models offers the possibility of comparing the protocols with each other based on the properties offered, implementation efficiency, etc. Starting from these, we propose that in the rest of this section, we identify the security and privacy classes in the HPVP model that fit the RFID services mentioned above.

Security. RFID security, which refers to unilateral or mutual identification and authentication, is studied against an adversary that can interact with the RFID system both impersonating tags and readers. The vast majority of security models also take into account the possibility of tag corruption by the adversary. A scheme that offers security against an adversary that can corrupt tags also offers security against one that cannot corrupt tags. However, the (algorithmic) costs of building corruption-resistant schemes are higher than in the case of schemes for which corruption resistance is not required. In addition, there are situations where resistance to corruption is an unjustified requirement. For example, the use of RFID tags for the identification and tracking of gloves and gowns in a hospital. It is unnecessary to demand that such tags be resistant to corruption.

As a result, and unlike the standard version of the HPVP model, we will consider two security classes:

1. *Weak security*, based on interaction with an adversary that cannot corrupt tags. *Weak unilateral identification and authentication* may be sufficient for ordinary medical assets. This category can include, for example, beds and wheelchairs for patients, protective equipment such as gloves, gowns, etc.
2. *Strong security*, or just *security*, based on interaction with an unrestricted adversary (that is, it can corrupt tags). For specialized medical assets, *strong mutual authentication* may be necessary to guarantee that the information from the tag is transmitted only to authenticated readers, in the presence of an adversary with

corruption capabilities. Medicines that are under strict control can enter this category. Also, the identification and tracking of patients and medical personnel fall into this category.

Privacy. The privacy property refers to links that can be established between the tags with which an adversary interacts and their identities. Anonymity, unlinkability, and traceability are examples of privacy properties. Any of these are important depending on the context in which they are used. What must be understood very well is that these properties are based on the link that the adversary can establish between the tags they interact with and their identities, and the type of interaction between the adversary and the tag matters a lot. According to these, we will consider the following privacy classes in the HPVP model as suitable for healthcare RFID:

1. *r-weak privacy.* This is weak privacy in the *randomized HPVP model* that is obtained from the HPVP model in which the oracle *Free(vtag)* is replaced by the oracle *r-Free(vtag)* as follows:

r-Free(vtag): the freed tag is re-randomized.

This means tags that accept a session without having the tag identifier randomized can be considered r-weak private. Disposable PPE, for example, can fall into this category. They are used only once, and as a result, their degree of privacy is very low. In fact, all disposable assets fall into this category;

2. *Weak privacy.* Some medical assets can be used multiple times, as well as between patients or medical staff. Even if they undergo a sanitization/sterilization process, it is sometimes useful not to make the route of the asset public among patients or medical staff. Considering that the corruption of the tag attached to the asset does not constitute a danger, the degree of privacy of these medical assets falls into the “weak privacy” class;
3. *Forward privacy.* This is the first class that considers tag corruption. The meaning of this privacy class is as follows. When the adversary tries to corrupt the tag of a medical device, equipment for monitoring medical procedures or maintaining patients’ life, it will emit a light or acoustic signal, alerting the medical staff. Thus, the adversary cannot continue his destructive activity, limiting himself only to concluding what he managed to obtain until that moment. The implants of a living patient can very well fall into this class;
4. *Destructive privacy.* The visitor’s card in a hospital allows the visitor access to certain hospital areas and to specific categories of patients. Such a card should offer a certain degree of privacy and a certain level of protection against corruption. In this case, the forward privacy level might not be suitable because corrupting a card would mean alerting the staff, causing panic (which can often be unjustified). The impossibility of using the card once corrupted would be an appropriate solution in this case. In other words, the technological process of the card should not allow its reuse once it has been corrupted, or the benefits that the adversary would obtain by impersonating a visitor should not justify the use of expensive corruption equipment that enables the reuse of the card;
5. *Strong privacy.* Access to facilities with significant restrictions (special operating rooms, organ storage locations for transplantation) or drugs with very restricted use requires strict control that must be resistant, even if the adversary can reuse corrupted cards (tags).

As it turns out, all privacy classes in the HPVP model naturally find their place in an RFID system for healthcare. The system architect, together with the healthcare institution’s management, will now decide the type of RFID subsystem used by service type.

4. Case Studies

In most cases, the first step of an RFID protocol is performed by the reader, who sends the tag a message that can include a freshly generated random. The tag’s answer may or may not depend on the reader’s initial query, but what is most important is that it must include a specific information to help the reader in the tag identification process. We call this particular information a *tag identifier*.

The non-randomization of the tag identifier leads to the total lack of privacy in the HPVP model, as shown below. To facilitate the discussion, we will assume that the tag identifier is sent to the reader in the tag’s first message (obviously, the debate can be extended).

Theorem 4.1. *Let Σ be a (PUF-based) RFID scheme for which it can be decided in deterministic polynomial time whether two tag's answers include or not the same tag identifier. If Σ has non-randomized tag identifiers, then it does not achieve narrow weak privacy in the HPVP model.*

Proof. (Sketch) Assume Σ is a (PUF-based) RFID scheme as in the theorem and the reader takes the first step in the protocol. Consider the following narrow weak adversary $\mathcal{A}$ that plays the following privacy game with Σ :

1. $\mathcal{A}$ creates two tags T_1 and T_2 and then draws (T_1, T_2) with some temporary identity $vtag_1$;
2. $\mathcal{A}$ plays with $vtag_1$ until it gets the first message that includes the tag identifier and then frees $vtag_1$;
3. $\mathcal{A}$ creates one more tag T_3 and then draws (T_1, T_3) with some temporary identity $vtag_2$;
4. $\mathcal{A}$ plays with $vtag_2$ until it gets the first message that includes the tag identifier;
5. $\mathcal{A}$ outputs 0 (the left privacy game) if the two messages from $vtag_1$ and $vtag_2$ include the same tag identifier, and 1, otherwise.

Step 5 above can be performed in deterministic polynomial time according to the theorem's hypothesis.

One can easily see that this adversary guesses with overwhelming probability which privacy game is playing. So, Σ does not achieve narrow weak privacy in the HPVP model. $\square$

Example 4.1. *Two protocols that fall in the hypothesis of Theorem 4.1 are in order:*

1. [16]: The tag sends in its first step, in the clear, the identifier SID ;
2. [10]: The tag sends in its first step, in the clear, the identifier IDS .

The use of temporary variables in the construction of protocols is generally necessary. In particular, global temporary variables have a vital role because they store values calculated at a specific step in the protocol to be checked in a later step. The HPVP model does not allow corruption of the virtual tag; it only provides corruption of the physical tag. As a result, to corrupt one of the two tags referenced by the virtual tag, the virtual tag must first be freed. The *Free* oracle resets precisely one of the two tags referenced by the virtual tag, which means that the values of the temporary variables are lost only for one of these tags. Obviously, this does not exclude the possibility that the adversary can identify the tag referred to by the virtual tag. The following general result applies exactly to this situation.

Theorem 4.2. *Let Σ be a (PUF-based) RFID scheme such that given two tags, given the sequence of messages exchanged by one of them with the reader in a protocol session until the tag has to authenticate the reader, and given the tags' full states, one can decide with overwhelming probability which of the two tags authenticates the reader. Then, Σ cannot achieve both reader authentication and narrow forward privacy in the HPVP model.*

Proof. (Sketch) Under the hypothesis of Theorem 4.2, we define the following narrow forward adversary $\mathcal{A}$:

1. $\mathcal{A}$ creates two tags T_1 and T_2 and then draws them under the temporary identity $vtag$;
2. $\mathcal{A}$ simulates the protocol between the reader and $vtag$ until the last protocol step played by reader (including this step) is reached and then frees $vtag$;
3. $\mathcal{A}$ corrupts both tags T_1 and T_2 and gets their full states;
4. With the reader's answer (previously obtained) and the tags full state's, $\mathcal{A}$ will deduce which of the two tags authenticates the reader;
5. If $\mathcal{A}$ decides that the reader is authenticated by T_1 , then it outputs 0; otherwise, it outputs 1.

One can easily see now that $\mathcal{A}$ can distinguish between the left and right privacy games with overwhelming probability. Thus, Σ cannot achieve both reader authentication and narrow forward privacy in the HPVP model. $\square$

Example 4.2. *The protocol in [16], already mentioned in Example 4.1, satisfies the requirements of Theorem 4.2. The tag uses the temporary variable N_T and also sends it in clear to the reader. So, the adversary, by corrupting the two tags drawn in the challenge phase can clearly decide which of them authenticated the reader.*

The same holds true for the protocols in [8, 10].

We note that the hypothesis of Theorem 4.2 does not explicitly call for the use of temporary variables. So, the result from Theorem 4.2 remains valid even if (global or not) temporary variables are not used.

Protecting temporary variables to avoid situations like those mentioned in Theorem 4.2 is crucial. The lack of space does not allow us to go into details on this problem, but we mention [17, 3] as primary steps taken in this direction.

We also say that the desynchronization between reader and tag is another major problem that often leads to protocols that reach an insufficient level of privacy [2].

5. Solutions and Challenges

In the previous sections, we have presented several protocols that have been proposed so far that do not ensure privacy in the HPVP model. Now the question arises: which RFID protocols satisfy the security and privacy requirements in the HPVP model, as they were enumerated in Section 3.2? Before answering this question we have to clarify an aspect. A protocol that achieves strong privacy will also accomplish the other levels of privacy. However, the primary interest for a particular intermediate level of privacy is to offer a protocol as efficient as possible. This will make the difference between, for example, a strong private protocol and a weak private one.

r-weak privacy. As far as we know, no RFID protocol proposed so far falls into this class. There is such a protocol for Vaudenay's privacy model [7]. We claim that this protocol is *r-weak private* in the HPVP model. We recall the protocol (Figure 2) but we skip the privacy proof due to the space constraint. The tag, which takes the first step in the protocol, computes $z = F_K(0, 0, x)$, where $F = (F_K)_K$ is a pseudo-random function, and sends it to the reader. The reader checks its database for a triple (ID, K, x) such that $z = F_K(0, 0, x)$ or $z = F_K(0, 0, x + 1)$. The reason is that at most one step of desynchronization may occur between reader and tag. When the reader finds the right value, resynchronizes with the tag and prepares the answer w . The tag checks the value w received from reader, takes a decision, updates x if necessary, and prepares the answer for reader. On receiving the tag's answer, the reader checks it, takes a decision, and updates x . If the reader does not update x (because it rejects the tag), then it will do so in the second step of the next protocol session (with the same tag). Therefore, the desynchronization between reader and tag is at most one step.

	Reader (DB, F)	Tag (K, x)
1		$\xleftarrow{z}$ $z = F_K(0, 0, x)$
2	If $\exists((ID, K, x) \in DB, i \in \{0, 1\}) : z = F_K(0, 0, x + i)$ then $x = x + i, w = F_K(0, 1, x + i)$ else $w \leftarrow \{0, 1\}^{\ell_2}$	$\xrightarrow{w}$
3		$w' = F_K(0, 1, x)$ If $w = w'$ then output $OK, x = x + 1, w' = F_K(1, 1, x)$ else output $\perp, w' = F_K(1, 0, x)$ $\xleftarrow{w'}$
4	If $w' = F_K(1, 1, x + 1)$ then output $ID, x = x + 1$ else output $\perp$	

Fig. 2. Stateful RFID scheme with constant tag identifiers that achieves *r-weak privacy* and reader-first authentication in the HPVP model

It is straightforward to check the correctness of the scheme in Figure 2. We remark that it does not use temporary variables, and the tag only needs to compute F . Moreover, the scheme allows for a quite efficient search procedure in the reader's database. Two ordered sets of indices are used: the first one with indices of the form $F_K(0, 0, x)$, and the second one with indices of the form $F_K(0, 0, x + 1)$. Both $F_K(0, 0, x)$ and $F_K(0, 0, x + 1)$ will point to the same database record (ID, K, x) . When z is received by reader, the first set of indices is searched for it; if not found, then the second set is searched for it. Therefore, it takes $O(\log n)$ time to search for a tag, where n is the size of the databases. When the reader authenticates the tag, x is updated both in the database and in the indices sets. Moreover, the indices sets must be resorted. This can be simply done in $O(\log n)$ time because the old indices have to be removed and the new ones have to be reinserted in the right position.

A more efficient search can be performed by using hash indices as in [1], where constant time is claimed for search. Moreover, the technique in [1] applied to our scheme works much better than for the scheme in [1]. This is because

the reader-tag desynchronization in our scheme is at most one step, while in [1] it is bounded by some polynomial $c(\lambda)$ in the security parameter λ . This fact leads to $c(\lambda)$ sets of indices in [1], while in our case we have only two.

Of course, the scheme is not weak private in the HPVP model because the tag identifier (z) is not randomized.

Weak privacy. The PRF protocol from [15] achieves unilateral authentication and weak privacy in the Vaudenay model but also in the HPVP model (see page 111 in [12]). In addition, the version from [13] achieves mutual authentication. Unfortunately, this protocol provides identification in linear time with respect to the number of tags in the database.

Forward privacy. Section VI of [6] proposes a forward-insider protocol as a version of the elliptic-curve-based Schnorr's identification scheme. The protocol uses public keys, which makes it somewhat inefficient. Searching the database consumes linear time with respect to its size. Designing a more efficient protocol that only covers forward(-insider) privacy remains an open problem.

Destructive and strong privacy. We do not know of any protocol that offers destructive but not strong privacy in the HPVP model. More generally, the relationship between strong and destructive privacy levels is not known in the HPVP model. Fortunately, the protocol in Figure 3 proposed in [4] ensures strong privacy in the HPVP model.

	Reader (DB, K_m)	Tag (ID, x, K'_m, P, s)
1		$K = P(s), K_m = K'_m \oplus K, x = x + 1, z = \{x \parallel ID\}_{K_m}$ $\xleftarrow{z}$ erase K, K_m
2	$\{z\}_{K_m^{-1}} = x \parallel ID$ If $\exists (ID, y) \in DB$ and $y < x$ then $y = x - 1, v \leftarrow \{0, 1\}^\ell, w = \{v \parallel (y + 1)\}_{K_m}$ else $w \leftarrow \{0, 1\}^{2\ell}$ $\xrightarrow{w}$	
3		$K = P(s), K_m = K'_m \oplus K, \{w\}_{K_m^{-1}} = v' \parallel x'$ If $x = x'$ then output $OK, x = x + 1, w' = \{x \parallel (v' + 1)\}_{K_m}$ else output $\perp, x = x + 1, w' = \{x \parallel (v' + 2)\}_{K_m}$ $\xleftarrow{w'}$ erase K, K_m
4	If $w' = \{(y + 2) \parallel (v + 1)\}_{K_m}$ then output $ID, y = y + 2$ else output $\perp$	

Fig. 3. Strong private and reader-first authentication RFID scheme in the HPVP model with temporary state disclosure

The scheme based on an symmetric-key cipher scheme $\{\cdot\}_{K_m}$ whose block-length is ℓ . The secret key K_m , also called the *master key* of the scheme, is stored on reader and all tags. However, to avoid key disclosure by tag corruption, K_m is masked by $K'_m = K_m \oplus P(s)$, where P is a PUF and s is a seed, both of them specific to the tag. The tag also stores its identity ID and a random value x needed to randomize the encryption. The value x is incremented each time a tag is queried. For the simplicity of the exposition we assume that ID, x, s , and $P(s)$ are all of the same length ℓ .

When the protocol session starts, the tag evaluates P on s , extracts K_m from K'_m , and sends its encrypted credentials (x, ID) to the reader. When the reader receives the message from tag, decrypts it and looks for a corresponding record in its database. If this is found, which means that the reader identified the tag, an “authentication code” w obtained from a random v and x is returned (remark that v is the first block to be encrypted). The tag decrypts w and checks the x -values. If they match, authenticates the reader and prepares an “authentication code” w' for reader. Note again that w' is built by placing x on the first position after it has been incremented. When receiving w' , the reader checks it against the value $\{(y + 2) \parallel (v + 1)\}_{K_m}$ computed by itself. If they match, the tag is authenticated and x is synchronized (by incrementing it) with the corresponding value on tag.

The RFID scheme in Figure 3 can be easily transformed into a weak private scheme. What we have to do is to remove the PUF from each tag and to keep the master key K_m in the tag's permanent memory. Although this might

seem a good idea from the constant-time identification point of view, this solution should be taken with great care because disclosure of K_m compromises the entire scheme.

There are other schemes that ensure strong privacy in the HPVP model, such as the PKE scheme from [15] (see also [6]). However, the above scheme does not use public key cryptography and ensures identification in constant time (the PKE scheme from [15] ensures identification in linear time).

6. Conclusion

The degree of adoption of RFID technology in healthcare is on the rise. The palette of healthcare services that can benefit from RFID systems is rich and not only limited to tracking medical equipment or people. We assist in incorporating this technology in tracking events and processes, thus increasing the degree of responsibility in its use. Security and privacy properties become extremely important.

Under these conditions, comparing RFID protocols using a security and privacy model is more than necessary. Also, the clear ranking of security and privacy needs by healthcare service classes helps the designer of RFID protocols a lot to offer both RFID protocols with suitable security and privacy properties but also efficient in practice.

Through this paper, we tried to answer the above problems. We highlighted the HPVP security and privacy model as suitable for studying RFID protocols and ranked healthcare services according to the degree of security and privacy required. We highlighted RFID protocols that can be used for each class. In addition, we presented two scenarios in which some of the RFID protocols proposed in the specialized literature fit but which, in the HPVP model, do not offer any privacy or mutual authentication. We see this work as the beginning of the systematization of studies on security and privacy needs in RFID technology (what will be) used in healthcare.

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